

AL and S methods: two extensions for L-method

Mário Antunes*, Henrique Aguiar†, Diogo Gomes*

*Instituto de Telecomunicações

Universidade de Aveiro

Aveiro, Portugal

Email: mario.antunes,dgomes@av.it.pt

†University of Oxford

Oxford, England

Email: henrique.aguiar@balliol.ox.ac.uk

Abstract—With the advent of smart IoT and M2M scenarios it becomes necessary to develop autonomous systems that optimize themselves with minimal human intervention. One possible method to achieve this is through Knee/elbow point estimation. Most of the time these points represent ideal compromises for parameters, methods and algorithms. However, estimating the knee/elbow point in curves is a challenging task. Our focus is on determining the ideal number of clusters autonomously. We analyse and discuss well-known knee/elbow estimators and two extensions based on the theoretical definition. The proposed methods (named AL and S methods) were evaluated against state-of-the-art estimators. The proposed methods are a viable stable solution for knee/elbow estimation.

Index Terms—Knee, Elbow, Optimization, Clustering, Unsupervised Learning

I. INTRODUCTION

Smarter IoT/M2M scenarios will require autonomous learning methods to cope with the complexity associated with the previously mentioned scenarios. As stated in [1] IoT/M2M will only achieve their full potential when the devices work and learn together with minimal human intervention. Thus, knee/elbow point estimation has become an important task for optimization. Generally speaking, the knee and elbow points represent the point where the performance increase and error minimization stagnates respectively. However, the lack of a formal definition makes it difficult to devise an efficient method. The conventional process to find it was to plot the graph and visually identify the point where the curve bends the most. Knee/elbow estimation becomes an inherently heuristic process since it requires a human user to manually identify the relevant point.

Nevertheless, knee/elbow estimation methods have been used in several areas. In fact, the concept of knee/elbow point in error curves is used in many fields like fatigue damage theories [2], detecting the number of clusters [3], botnet detection [4], and system behaviour [5]. With the advent of the Internet of Things (IoT) [6], Machine to Machine (M2M) [7] communications and Machine Learning (ML) [8] several scenarios require a system that works autonomously with minimal human intervention. As such, optimization

methods based on knee/elbow point estimation has become an important research area.

Our personal interest comes from clustering distributional profiles of words into categories in order to improve the accuracy of semantic similarity [1]. We developed an unsupervised method to learn a semantic model from public web services. The learning method uses clustering algorithms to identify word categories automatically (word category is closely related to concepts and the possible meaning of a word). Our prototype relied on k -means to cluster the distributional profile into categories. However, the algorithm has two main disadvantages: it is not deterministic and requires the number of clusters *a priori*. At the moment we rely on a statistical method [9] (similar to GAP statistics [10]) to determine the ideal number of clusters. Although the results were positive, we are interested in evaluating and researching other means to determine the ideal number of clusters.

One of the main difficulties for cluster analysis is to determine the correct number of clusters for different datasets. Most clustering algorithms are designed only to partition data objects according to a known number of clusters. As such, identifying the correct number of clusters is an important task for any clustering problem. Especially important when considering complete autonomous systems and unsupervised learning methods. There are several methods to estimate the optimal number of cluster [11]–[13], nevertheless these methods normally require complex computations or some prior knowledge regarding the input data. As an alternative, some methods rely on identifying the knee/elbow point in performance curves and select it as the ideal number of clusters. Reliable algorithms to estimate the knee/elbow point are a necessity to use the previously mentioned methods. In previous works we explored this topic [14], formalized to some extent the concept of knee/elbow and proposed a novel algorithm based on the first derivative dynamic threshold. In this paper, and based on the previous formalization, we propose two extensions for the well-known L-method [15].

The remainder of the paper is organized as follows. In Section II we review the concept of knee/elbow as the point of maximum curvature. An overview of discrete knee/elbow

estimation algorithms is given in Section III. Our proposed methods are described in Section IV. The results of our evaluation are given in Section V. Finally, discussion and conclusions are presented in Section VI.

II. KNEE/ELBOW POINT

We tried to formalize this concept in a previous publication [14]. Nevertheless, we present here a short summary of the most relevant points. As previously stated, knee/elbow estimation is an inherently heuristic process. However, to design a knee/elbow estimation algorithm, we require a consistent definition.

The authors [5], [15] used the mathematical definition of curvature for continuous functions as the basis for knee/elbow definition. For any continuous function $f(x)$ there exists a closed-form $K_f(x)$ that defines the curvature of $f(x)$ at any given point:

$$K_f(x) = \frac{f''(x)}{(1 + f'(x)^2)^{\frac{3}{2}}} \quad (1)$$

Based on this definition the knee/elbow point is the point of maximum curvature since curvature is a mathematical measure of how much a function differs from a straight line. We can find maximum curvature by calculating the derivative of Equation 1 and finding its roots. As depicted in Equation 2, it is important to mention that we discarded the denominator. The denominator is not relevant, for finding the roots, because it cannot be equal to zero (otherwise the function tends to infinity). The roots of a fractional function are the roots of the numerator, the denominator is only relevant to verify that the numerator roots are not also denominator roots (which makes the function tend to infinity).

$$K'_f(x) = f'''(x)(1 + f'(x)^2) - 3f''(x)^2 f'(x) = 0 \quad (2)$$

From the previous equations, we can derive some important pieces of information. First, inflexion points (or even the first and second derivative by itself) are not sufficient criterion to reliably identify knee/elbow. In contrast, the curvature definition precisely matches the concept of a knee. Figure 1 and Figure 2 depicts the function $f(x) = \frac{1}{x}$ and $f(x) = \ln(x)$. The respective curvature functions are expressed in Equation 3 and Equation 4 respectively.

$$K_{\frac{1}{x}}(x) = \frac{2}{x^3(1 + x^{-4})^{\frac{3}{2}}} \quad (3)$$

$$K_{\ln(x)}(x) = -\frac{1}{x^2(1 + x^{-2})^{\frac{3}{2}}} \quad (4)$$

Although $f(x) = \frac{1}{x}$ first and second derivatives do not have any roots, according to the curvature function $K_f(\frac{1}{x})$ the maximum curvature is 1. Similarly, $f(x) = \ln(x)$ first and second derivatives also do not have any roots, again according to the curvature function $K_f(\ln(x))$ the maximum curvature is around 0.7.

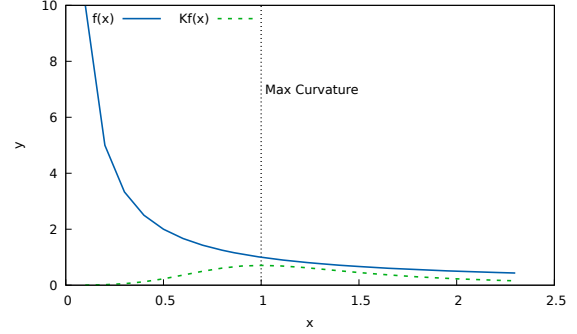


Fig. 1: Graphical representation of $f(x) = \frac{1}{x}$, the respective curvature $K_{\frac{1}{x}}(x) = \frac{2}{x^3(1+x^{-4})^{\frac{3}{2}}}$ and the point with maximum curvature 1

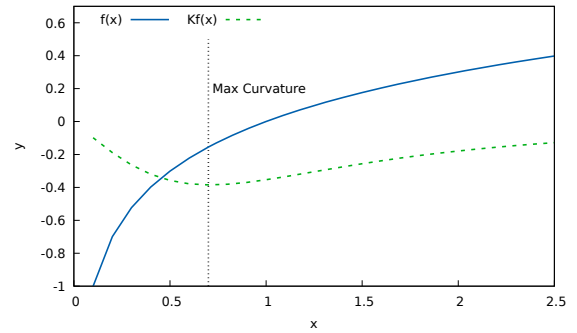


Fig. 2: Graphical representation of $f(x) = \ln(x)$, the respective curvature $K_{\ln(x)}(x) = -\frac{1}{x^2(1+x^{-2})^{\frac{3}{2}}}$ and the point with maximum curvature 0.7

Second, rescaling the function $f(x)$ alters the value of the knee/elbow point. A rescaling operation can be expressed as multiplication and addition of constants: $af(x)+b$, where a and b are constants. Applying the previous results to the curvature equation we end up with Equation 5.

$$K_{af+b}(x) = \frac{af''(x)}{(1 + a^2f'(x)^2)^{\frac{3}{2}}} \quad (5)$$

Again, in order to identify the maximum curvature, we can calculate the derivative of Equation 5 and find its roots. Equation 6 is only similar to the original equation (Equation 2) when $a = 1$. In short, rescaling the original function $f(x)$ may alter the curvature of the function, and impact the detection of the knee/elbow point.

$$K'_f(x) = f'''(x)(1 + a^2f'(x)^2) - 3a^2f''(x)^2 f'(x) = 0 \quad (6)$$

While curvature is well-defined for continuous functions, the same is not true for discrete datasets. In a previous work [14] we tried to fit a continuous function to the discrete dataset and use Equation 2 to identify the point with the highest curvature. Unfortunately, our evaluation showed that this method produced poor results. Fitting a continuous function to a set of arbitrary data points

is difficult (especially for non-linear functions). As such, several researchers have developed methods to estimate the knee/elbow point without the close-form expressed in Equation 1.

III. BACKGROUND AND RELATED WORK

Although not heavily researched, there are several approaches to detecting knee/elbow points in discrete data. In this section, we describe the three most common approaches.

Kneedle [5] uses the point further away from a line defined by the head and tails points of the error curve. Both coordinates of the original curve are rescaled to $[0, 1]$, in order to easily find the point with maximum curvature. It is important to notice that the method supports detection of knee/elbow points online. The authors use this feature to detect relevant points in congestion control and network latency. Nonetheless, this method suffers from two drawbacks. First, if the tail of the error curve is long enough it may alter the knee/elbow point estimation. Second, the method applies rescaling to the original error curve. As seen in Section II, rescaling operations may alter the curvature of the function. These drawbacks are most evident in offline scenarios.

L-method [15] fits two straight lines from the head to candidate point and from the candidate point to the tail of the curve, hence the name **L**. The candidate point that minimizes the Root Mean Squared Error (RMSE) between the straight lines and the points of the curve is returned as the knee/elbow point (see Figure 3). The resulting point represents the point with the sharper angle in the curve. A long tail in the error curve can also alter the value of the knee/elbow point with this method. To minimize this issue the authors proposed an iterative method that incrementally reduces the tail, while the knee/elbow point is refined. The major drawback of this method is the high computational cost necessary to fit the straight lines.

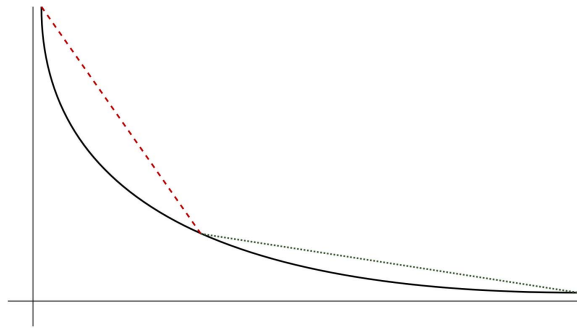


Fig. 3: Graphical representation of the *L*-method.

DFDT [14] is a hybrid method. It estimates the first derivative of the curve. The first derivative of the curve represents the slope of the tangent line. Similar to the *L*-method, this method tries to identify the point where the function was a sharp angle. Instead of fitting two straight

lines, the method uses a threshold algorithm to find the ideal slip between high and low values of the first derivative. Dividing the slopes into two classes, the head of the curve with high values and tail with low values. The point closer to the threshold is used as the knee/elbow estimation.

IV. AL AND S METHODS

As initially stated, our main interest in knee/elbow point estimation is due to the need to autonomously identify the ideal number of clusters. In a previous publication, we evaluated the impact of clustering in semantic models [16] and the quality of the clustering is directly correlated with the quality of the model. In several clustering algorithms, the method to identify the ideal parameters is finding a compromise in an error curve (typically with a sharp curvature). *K*-means [17] uses a method properly named elbow method. DBSCAN [18] uses the elbow of a *k*-distance graph (plotting the average distance to the *k* nearest neighbours ordered from the largest to the smallest value [19]) as the ideal value for ϵ . Hierarchical clustering relies on knee/elbow detection to select the ideal cluster from the dendrogram.

As stated previously, the formal definition of curvature cannot be used directly in discrete sets of points. We have tried a regression-based method, but it performed poorly. Nevertheless, the formal definition can provide some insights, useful to implement a better model.

In this work, we propose two different methods, that are an extension of the popular *L*-method [15]. The first one is named *AL*-method and it combines the conventional *L*-method score with another based on the angle between the two fitted lines. The second method, named *S*-method, fits three lines instead of two. In the following subsections, we describe in details the previously mentioned methods.

A. *AL*-method

The main idea for this method comes directly from the formal definition. Knee/elbow point is the point with maximum curvature, which means the point where the curve differs the most from a straight line.

One way to achieve this is to compute the angle α between the lines fitted by the conventional *L*-method (see Figure 4) using the following equation, where m denotes the slope of each line:

$$\alpha = \arctan \frac{m_1 - m_2}{1 - m_1 m_2} \quad (7)$$

The angle closer to 90 degrees (right angle) represents the sharper bend in the curve. The angle score is computed with the following expression:

$$score_i = |90 - \alpha_i|^2 \quad (8)$$

In the conventional *L*-method select the point that minimizes the Root Mean Squared Error (RMSE) between the fitted lines. This score is combined with the angle α between the lines (see Equation 8) after rescaling both

scores to the range of $[0, 1]$. The point with the minimum combined score is selected as the knee/elbow point.

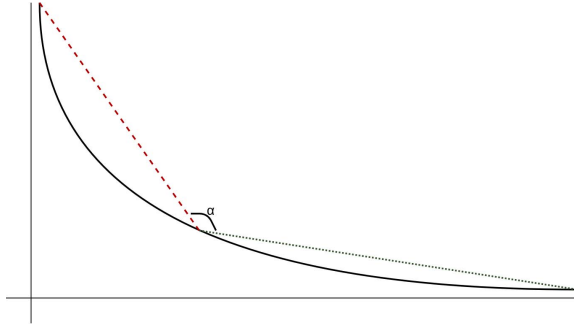


Fig. 4: Graphical representation of the AL-method.

The algorithm is described in detail in Algorithm 1. It is important to notice that we also added the iterative refinement used in the conventional L -method. For the evaluation (see Section V) we also considered a method that only relied on the angle score (named A -method).

Algorithm 1 AL-method

```

1: function ALMETHOD( $x, y, length$ )
2:    $idx \leftarrow 1$ 
3:    $lrl \leftarrow regression(x, y, 0, 0, idx + 1)$ 
4:    $lrr \leftarrow regression(x, y, idx, idx, length - idx)$ 
5:    $lmetrics \leftarrow [length - 2]$ 
6:    $ametrics \leftarrow [length - 2]$ 
7:    $lmetrics[0] \leftarrow L - score(lrl, lrr)$ 
8:    $ametrics[0] \leftarrow A - score(lrl, lrr)$ 
9:   for  $i \leftarrow 2; i < length - 1; i++$  do
10:     $lrl \leftarrow regression(x, y, 0, 0, i + 1)$ 
11:     $lrr \leftarrow regression(x, y, i, i, length - i)$ 
12:     $lmetrics[i - 1] \leftarrow L - score(lrl, lrr)$ 
13:     $ametrics[i - 1] \leftarrow A - score(lrl, lrr)$ 
14:   end for
15:    $lmetrics \leftarrow rescaling(lmetrics)$ 
16:    $ametrics \leftarrow rescaling(ametrics)$ 
17:    $score \leftarrow add(lmetrics, ametrics)$ 
18:   return  $min(score) + 1$ 
19: end function

```

B. S-method

One of the drawback of these methods is dealing with long head/tails. When the curve has a high slope in the first points, that area is considered a long head. On the other hand, if the end of a curve as a slope close to zero, that area is named a long tail. These areas produce bias on the knee/elbow point estimation.

The second proposed algorithm is named S-method and was designed to minimize this issue. Instead of fitting two straight lines in a curve, it fits three lines. The rationale behind this idea is the following: the first and third lines are fitted on the long head/tail respectively, the second

line is fitted in between. The second line should capture the correct nature of the curve (see Figure 5).

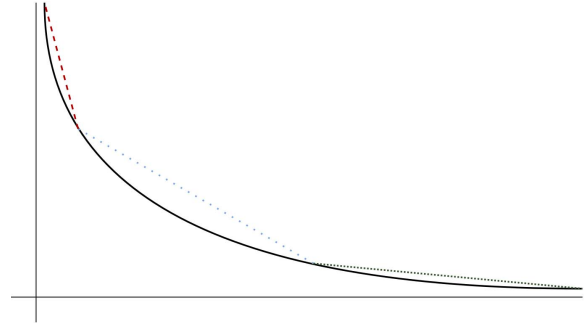


Fig. 5: Graphical representation of the S-method.

The conventional L -method produce a single point as output, the S -method produces two points. The first point is used for knee estimation, while the second is used for elbow estimation. This statement was evaluated through empirical validation.

The algorithm is described in detail in Algorithm 2. Similarly, to the previous method, we also added the iterative refinement used in the conventional L -method.

Algorithm 2 S-method

```

1: function SMETHOD( $x, y, length$ )
2:    $p \leftarrow 1, 2$ 
3:    $smetric \leftarrow PositiveInfinity$ 
4:   for  $i \leftarrow 1; i < length - 2; i++$  do
5:      $lrl \leftarrow regression(x, y, 0, 0, i + 1)$ 
6:     for  $j \leftarrow i + 1; j < length - 1; j++$  do
7:        $lrm \leftarrow regression(x, y, i, i, j - i)$ 
8:        $lrr \leftarrow regression(x, y, j, j, length - j)$ 
9:        $rmse \leftarrow S - score(lrl, lrm, lrr)$ 
10:      if  $rmse < smetric$  then
11:         $p[0] \leftarrow i$ 
12:         $p[1] \leftarrow j$ 
13:         $smetric \leftarrow rmse$ 
14:      end if
15:    end for
16:  end for
17:  return  $p$ 
18: end function

```

Finally, it is important to mention the major drawbacks of this method. The first one is the execution time, the algorithm is considerably slower than the L -method. Computing the ideal regression of three straight lines requires an extra loop.

The proposed methods were developed in Java and are public available¹. Furthermore, all the other methods mentioned in Section III were also implemented and are also publicly available in the same library. The code is

¹<https://github.com/mariolpantunes/ml>

integrated into a small ml library that is currently used for research purposes.

V. PERFORMANCE EVALUATION

In order to evaluate our method, we selected five datasets from UCI Machine Learning Repository² and three datasets from Clustering Basic Benchmark [20].

The five datasets from UCI Machine Learning Repository were: Iris, Yeast, Control, Wine and Vehicle. **Iris** contains three classes of 50 instances each, where each class refers to a type of iris plant. **Yeast** is originally used for protein localization sites prediction. The class distribution from a rule-based expert system indicates the optimum number of clusters as 10. However, as the size of six clusters among them is too small, we considered 5 as the optimal number of clusters. **Control** contains 600 examples of control charts synthetically generated. There are six different classes of control charts. **Wine** contains 178 data points that are the results of a chemical analysis of wines grown in the same region in Italy but derived from three different cultivars. All data points are composed of 13 attributes, divided into 3 classes. Finally, **Vehicle** contains 946 data points that represent vehicle silhouettes. All data points are composed of 18 attributes, divided into 4 classes.

The three datasets from Clustering Basic Benchmark are named **A-sets** (A1, A2 and A3). They are synthetic 2d data with a varying number of vectors and clusters. There are 150 vectors per cluster.

We clustered the eight datasets with *K*-means algorithm and SLINK [21] (Hierarchical Clustering) and used the previously mentioned knee/elbow estimation methods to identify the ideal number of clusters. The ranges used to cluster the datasets were related to the expected number of clusters: E is the expected number of clusters thus the range was defined as $[E/2, E \times 2]$. The dynamic range provides insight into how the number of sample points impacts the estimation of the knee/elbow point. In order to evaluate the performance of each method, we computed the RMSE between the correct and the estimated number of clusters.

Table I and Table II contains the result of our evaluation for *K*-means and SLINK respectively.

Considering the *K*-means clustering algorithm three methods were extremely accurate: Kneedle, DFDT and *S*-method. On the other hand, the *A* and *AL* performed poorly. This result show that the angle between the fitted straight lines is not sufficient, or does not add useful information to improve the knee/elbow point estimation. One possible reason for this result is how the regression is computed. The straight lines were computed to fit all the points in the selected range (linear regression), instead of simply using the first and last point in the range. By using all the points in the range we introduce a bias on the slope of the linear regression. In the future, we intend

to explore this issue and evaluate an improved version of the previously mentioned method.

The methods that performed well with the SLINK algorithm were: Kneedle and *L*-method. We can assume that the underlying clustering algorithm has some impact on the accuracy of the *L*-method. Take into account that the previously mentioned method was developed for hierarchical clustering. Furthermore, this also shows that the underlying clustering algorithm has some impact on the shape of the curve. This explains how the same knee/elbow methods perform so differently.

Finally, all the methods performed well for small values of clusters (the first five datasets). When the number of expected clusters grew, the accuracy of the methods tends to degrade. All the methods mentioned in the work require a certain number of points before and after the knee/elbow point in order to detect it. For this evaluation, we centred the expected value in the range, but for real-world scenarios, this information may not be available. Further research is required to fully understand this issue, evaluate other scenarios and discover the limitations of the algorithms.

VI. DISCUSSION AND CONCLUSIONS

Estimating the knee/elbow point in error curves is an open challenge. Nevertheless, these points represent the ideal compromises for several parameters, methods and algorithms. With the advent of smart IoT, M2M scenario, autonomous optimization becomes an important requirement. Due to this, knee/elbow estimation has become an important topic. As previously stated, our focus is on determining the ideal number of clusters for our semantic model [1].

In this paper, we proposed two extensions for the well-known *L*-method. The first one, *AL*-method, enhances the *L*-method score with the angle between the fitted straight lines. The second one, *S*-method, fits three straight lines instead of only two.

The previously mentioned methods were evaluated against the state of the art. The evaluation showed that the second proposed method is a stable viable option for estimation knee/elbow points in curves. Further research is necessary to comprehend the limitations of the algorithm and improve the performance of the implementation. On the hand, the first method performed poorly. This result shows that the angle between the fitted lines did not contribute positively to the point estimation. One possible reason for this is due to the regression used by the *L*-method. As such, the proposed method should be researched further.

We intend to continue this research line, improve our understanding of the methods and comprehend the limitations of each one. One interesting result is that the underlying clustering algorithm introduces some bias on the curve. This explains why the same point estimation methods performed so differently when the clustering algorithm is different. In order to identify the limitation of the

²<https://archive.ics.uci.edu/ml/index.php>

TABLE I: Results of the evaluation with the K -means clustering algorithm.

Dataset/ Method	Iris 3	Yeast 10	Control 6	Wine 4	Vehicle 4	A1 20	A2 35	A3 50	RMSE
Kneedle	4	10	6	4	5	18	32	44	2.5
L -method	3	10	6	3	4	18	27	37	5.4
DFDT	4	10	6	4	4	20	39	46	2.1
S-method	4	12	7	4	5	17	32	54	2.3
A-method	3	10	11	5	7	39	68	97	23.9
AL-method	3	10	7	5	6	34	51	54	11.9

TABLE II: Results of the evaluation with the SLINK clustering algorithm.

Dataset/ Method	Iris 3	Yeast 10	Control 6	Wine 4	Vehicle 4	A1 20	A2 35	A3 50	RMSE
Kneedle	4	10	7	4	4	18	32	48	1.6
L -method	3	10	6	4	4	17	39	47	2.1
DFDT	4	8	6	4	4	12	27	44	4.6
S-method	4	12	8	4	5	20	39	71	7.6
A-method	5	12	11	4	5	39	68	95	22.3
AL-method	4	11	7	4	5	20	39	71	8.6

proposed method, a more thorough evaluation is required. Considering other clustering algorithms, synthetic datasets, scenarios and applications. We are preparing one evaluation based on our semantic model.

ACKNOWLEDGEMENT

The present study was developed in the scope of the **Smart Green Homes** Project [POCI-01-0247-FEDER-007678], a co-promotion between **Bosch Termotecnologia S.A.** and the **University of Aveiro**. It is financed by Portugal 2020 under the Competitiveness and Internationalization Operational Program, and by the European Regional Development Fund.

REFERENCES

- [1] M. Antunes, D. Gomes, and R. L. Aguiar, "Towards IoT data classification through semantic features," *Future Generation Computer Systems*, dec 2017.
- [2] L. Franke, "A non-linear fatigue damage rule with an exponent based on a crack growth boundary condition," *International Journal of Fatigue*, vol. 21, no. 8, pp. 761–767, sep 1999.
- [3] Q. Zhao, V. Hautamaki, and P. Fränti, "Knee point detection in BIC for detecting the number of clusters," in *Advanced Concepts for Intelligent Vision Systems*. Springer Berlin Heidelberg, 2008, pp. 664–673.
- [4] A. Karasaris, B. Rexroad, and D. Hoefflin, "Wide-scale botnet detection and characterization," in *Proceedings of the First Conference on First Workshop on Hot Topics in Understanding Botnets*, ser. HotBots'07. Berkeley, CA, USA: USENIX Association, 2007, pp. 7–7. [Online]. Available: <http://dl.acm.org/citation.cfm?id=1323128.1323135>
- [5] V. Satopaa, J. Albrecht, D. Irwin, and B. Raghavan, "Finding a 'kneedle' in a haystack: Detecting knee points in system behavior," in *2011 31st International Conference on Distributed Computing Systems Workshops*. IEEE, jun 2011.
- [6] F. Wortmann, K. Flüchter *et al.*, "Internet of things," *Business & Information Systems Engineering*, vol. 57, no. 3, pp. 221–224, 2015.
- [7] K.-C. Chen and S.-Y. Lien, "Machine-to-machine communications: Technologies and challenges," *Ad Hoc Networks*, vol. 18, pp. 3–23, 2014.
- [8] V. N. Vapnik, "An overview of statistical learning theory," *IEEE Transactions on Neural Networks*, vol. 10, no. 5, pp. 988–999, 1999.
- [9] D. T. Pham, S. S. Dimov, and C. D. Nguyen, "Selection of k in k -means clustering," *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, vol. 219, no. 1, pp. 103–119, 2005.
- [10] R. Tibshirani, G. Walther, and T. Hastie, "Estimating the number of clusters in a data set via the gap statistic," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 63, no. 2, pp. 411–423, 2001.
- [11] B. J. Ross, "A comparison of knee strategies for hierarchical spatial clustering," in *Lecture Notes in Computer Science*. Springer International Publishing, 2018, pp. 81–87.
- [12] S.-S. Yu, S.-W. Chu, C.-M. Wang, Y.-K. Chan, and T.-C. Chang, "Two improved k -means algorithms," *Applied Soft Computing*, vol. 68, pp. 747–755, jul 2018.
- [13] R. Ünlü and P. Xanthopoulos, "Estimating the number of clusters in a dataset via consensus clustering," *Expert Systems with Applications*, vol. 125, pp. 33–39, jul 2019.
- [14] M. Antunes, J. Ribeiro, D. Gomes, and R. L. Aguiar, "Knee/elbow point estimation through thresholding," in *2018 IEEE 6th International Conference on Future Internet of Things and Cloud (FiCloud)*. IEEE, aug 2018.
- [15] S. Salvador and P. Chan, "Determining the number of clusters/segments in hierarchical clustering/segmentation algorithms," in *16th IEEE International Conference on Tools with Artificial Intelligence*. IEEE Comput. Soc, 2004.
- [16] M. Antunes, D. Gomes, and R. L. Aguiar, "The impact of clustering for learning semantic categories," in *Proceedings of the 3rd International Conference on Internet of Things, Big Data and Security*. SCITEPRESS - Science and Technology Publications, 2018.
- [17] S. Lloyd, "Least squares quantization in pcm," *Information Theory, IEEE Transactions on*, vol. 28, no. 2, pp. 129–137, 1982.
- [18] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters a density-based algorithm for discovering clusters in large spatial databases with noise," in *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining*, ser. KDD'96. AAAI Press, 1996, pp. 226–231. [Online]. Available: <http://dl.acm.org/citation.cfm?id=3001460.3001507>
- [19] E. Schubert, J. Sander, M. Ester, H. P. Kriegel, and X. Xu, "DBSCAN revisited, revisited," *ACM Transactions on Database Systems*, vol. 42, no. 3, pp. 1–21, jul 2017.
- [20] P. F. *et al*, "Clustering basic benchmark," 2015. [Online]. Available: <http://cs.uef.fi/sipu/datasets/>
- [21] R. Sibson, "Slink: an optimally efficient algorithm for the single-link cluster method," *The computer journal*, vol. 16, no. 1, pp. 30–34, 1973.